
How much search budget a target fidelity actually needs

Answering last checkpoint's action item: two boundary runs, decided on validation only.

Background: the previous checkpoint showed semantic proposals beat random and greedy search, but ranked methods at one fixed budget.

Question: what is the smallest candidate budget that reaches a target validation fidelity in 10 of 12 seeds?

Method: re-analysed 7 earlier conditions with no model calls, then ran 2 pre-registered boundary cells; judged on validation only.

Result: the open loop passes at budget 6 (12/12) where the closed loop fails (8/12); more budget did not rescue the XXZ cell (9/12 to 5/12).

Not claimed: no single minimum budget and no interval - the counts are not monotone; the test set was never evaluated.

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Project: llm-vqc · Previous checkpoint: one-factor robustness stress test · New model calls: 138 · candidate circuits trained: 408 · Measured cost at list price: USD 0.29 (hard cap USD 2.00, authorised in advance)

llm-vqc

What budget a required accuracy needs, not who wins

Every earlier comparison ranked methods at one fixed budget; that is not the operational question.

For a target validation trash fidelity, what is the smallest candidate budget that reaches it in at least 10 of the same 12 paired seeds?

- **Claim under test (EXPERIMENT_CHECK_REQUIRED before this run): a budget below 8 suffices on the Ising chain, and the XXZ cell that fell one seed short at budget 8 is rescued by more budget.**
- Alternative that fits the same evidence (MODEL_INFERENCE): attainment counts near a threshold are dominated by run-to-run variation, so a budget change moves the count without moving the underlying fidelity distribution much.
- Both were separable in advance, because each budget is run as its own independent policy rather than as a prefix of a larger run.
- A result could exist under a false claim only if the counts were monotone in budget by construction. They are not: the runs are independent, so this was a genuine test.

The endpoint is a count of seeds over a fixed threshold, not a mean fidelity.

Next: What was already settled, and what this adds.

Settled: semantics helps. Open: how much search

Position relative to the immediately preceding checkpoint of this project.

Settled (measured)

- Beat non-semantic search in all seven tested conditions.
- Survives changes of budget, width, Hamiltonian and model.

Shown fragile (measured)

- The closed-loop bonus is not general; it moves with every factor tested.
- Invalid proposals become random draws.

The gap this run addresses

- No budget between 4 and 8 had ever been run.
- One cell sat one seed short and was never retried at a larger budget.

Novelty: recombined

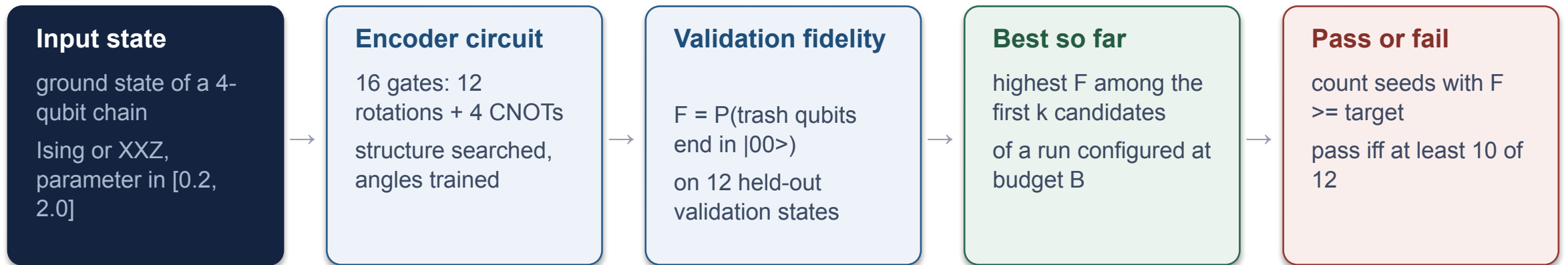
- Methods, prompts, trainer and seeds unchanged; only the budget moves.
- Closest prior work: our own last checkpoint, reused as controls.

Next: Every term the rest of the deck uses.

From a ground state to a single pass-or-fail count

Each arrow is a fixed rule, written down before the runs; nothing below was chosen after seeing a result.

Budget B candidate circuits trained and evaluated per seed - not tokens, not model calls, not wall-clock time



A mean above the target is not a pass, and equality with the target counts as a hit.

Next: The judging rules, and the one that was deliberately changed.

The judging rules were frozen before any run

Fixed in the repository before the first new candidate circuit was generated.

Rule	Fixed as
Endpoint	Validation trash fidelity; never test
Targets	0.95 and 0.99, both reported, neither revisable
Seeds	The same 12 paired seeds, 0-11, all reported
Pass rule	At least 10 of 12; equality counts
Admissible budgets	Even budgets 4, 6, 8, 10, 16 only
Budget split	Half exploration, half refinement, always
Inherited unchanged	Model, prompts, schema, repair policy, gates, trainer, splits, selection
Cost	Refuses to call without an explicit cap; USD 2.00

Reported as	Verified pass / verified fail / unverified
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Does not show: The table does not show effect sizes or significance tests; the endpoint here is a count, and no paired statistic is claimed from it.

- One deliberate change from the action item: no interval for the minimum is reported.
- An interval would assume attainment never disappears as the budget grows. That assumption is testable, and slide 7 shows it failing, so the three-way report replaces it.

Next: What was actually executed, and what was reused.

Two new runs, seven reused, no test evaluation

Existing logs were re-analysed first; new calls were made only at the two unresolved boundaries.

Reused (no model calls)

- Seven earlier conditions, 2,880 candidate records.
- Reproduces the earlier audit byte for byte (CONFIRMED).

Run now

- Ising chain at budget 6, all four methods, same seeds.
- XXZ chain at budget 10, closed loop only, same seeds.
- 408 candidate circuits trained and evaluated.

Checks that passed

- Seed completeness, candidate order, finite fidelity.
- A machine check refuses any two-factor cell.
- 603 automated tests pass, 46 of them new.

Not verified

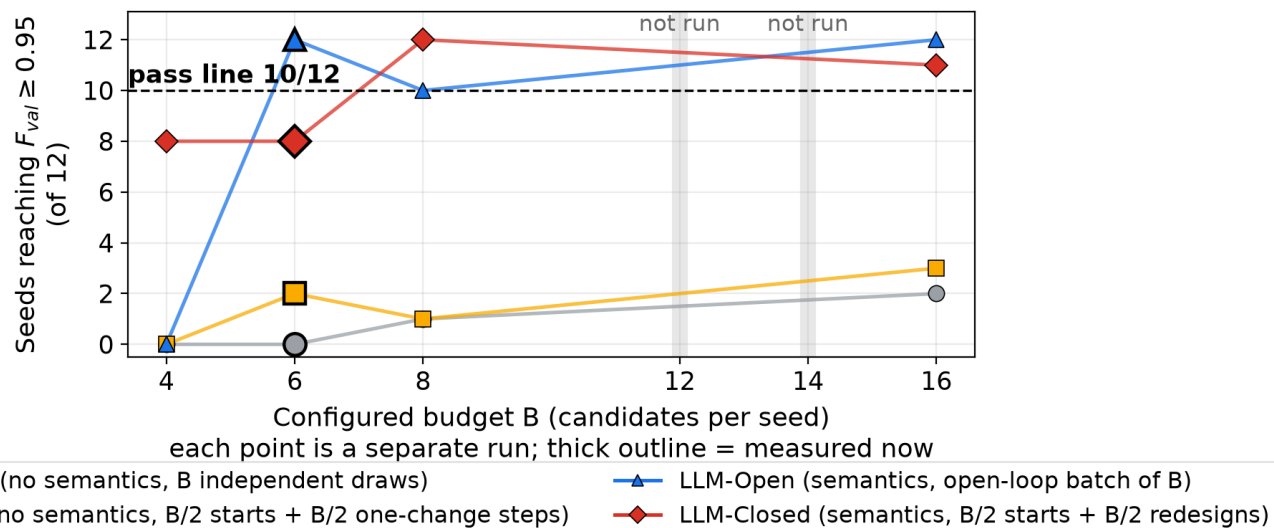
- No confirmatory evaluation on any held-out set exists, by design.
- One run was interrupted; the resume paid nothing for stored seeds.

Next: The main evidence.

Runner, analysis, figures, report and this deck are generated by checked-in scripts; the reproduction commands are in the report.

Open clears the bar at a budget closed cannot

Ising chain, target 0.95, four independently executed budgets, the same 12 paired seeds.



Metric: validation trash fidelity (never test). 12 paired seeds (0-11). Pass = at least 10/12 seeds at or above the target. Runs at different B are separate policies and are never concatenated.

x: budget B. y: seeds (of 12) reaching validation fidelity 0.95. Dashed: the 10-of-12 rule. Thick outline: measured here; grey band: not run.

Does not show: A count, not an average, so no error bars. Nothing about budgets 12 or 14, other widths, or the test set.

Next: What this does and does not mean.

Measured

- Open 12/12 passes, closed 8/12 fails.
- Random 0/12, greedy 2/12, same seeds.

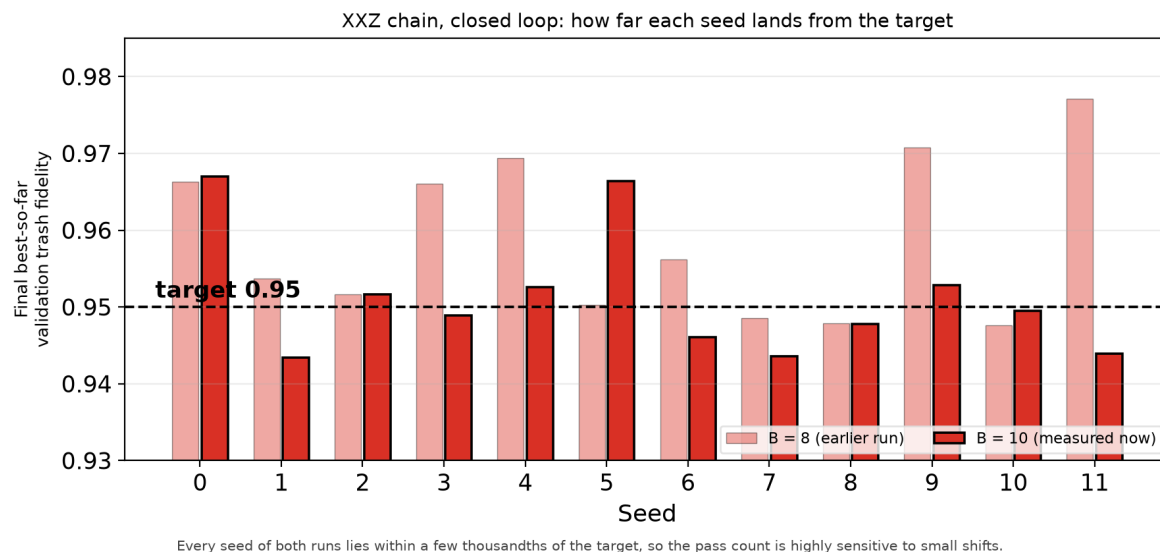
Counts are not monotone

- Open at B=4,6,8,16: 0, 12, 10, 12.
- Closed: 8, 8, 12, 11.

Generated from the committed validation-only result tables by the checked-in figure script.

The XXZ count halved while the fidelities barely moved

Closed loop at the XXZ anchor, budget 8 against budget 10, per seed, same 12 paired seeds.



x: seed. y: that seed's final best validation fidelity - pale for budget 8, solid for the budget 10 measured here. Dashed line: the 0.95 target. Single values, not averages.

Does not show: Not the path within a run, the rejected candidates, or whether a third run lands the same way.

Observed (measured)

- Seeds at target: 9 to 5 of 12.
- Mean moved only -0.0076; 17 of 24 within 0.01.

Interpretation (inferred)

- A threshold inside the distribution makes the count knife-edge.
- Only the budget varied: no cause isolated.

Supports: the budget boundary is real and the count is threshold-sensitive. **Does not support:** any causal story about feedback.

Next: What would have to be true for this to be wrong.

What bounds this, and what stays unresolved

Stated before any of it is used to argue for a next step.

What would overturn it

- A repeat landing the other side of the line - margins are thousandths.
- A different target, which would reshuffle every count.

Validity range

- 4 qubits, noiseless, one capacity rule, one model, one XXZ anchor.
- One open-loop pool shared across the 12 seeds, not 12 generations.

Negative results kept

- The closed loop fails the rule at budget 6.
- The XXZ boundary probe did not rescue the cell.
- Target 0.99: 0 of 12 everywhere, unresolved.

Left open on purpose

- 6 and 8 qubits and the other model: every method sits at 0-3 of 12.
- A blind sweep cannot separate the causes.
- Random fallbacks stay in the score: 12 of 408 evaluated candidates.

Open question: Attainment counts are not monotone in budget, so no statement of the form 'the minimum lies between X and Y' is supported by these runs.

Next: What needs deciding.

Sharpen the boundary, or diagnose the 0.99 ceiling

Both new cells are complete; the next move needs a decision, not more of the same.

What the evidence supports

- **Ising chain, target 0.95: smallest verified passing budget is 6 for the open loop, 8 for the closed loop; budget 4 fails for both.**
- XXZ chain, closed loop: no verified passing budget; 8 and 10 are both verified failures (measured).
- Target 0.99 unresolved everywhere, both new cells included (null result).
- New work: 138 model calls, USD 0.29 at list prices, under a USD 2.00 cap.

Decisions needed from you

- **D1 Sharpen the budget boundary?**
 - (a) repeat both cells with fresh generation seeds
 - (b) accept budget 6 open-loop as the answer
 - Recommend (a), Ising only; moderate.
- **D2 What to do about 0.99**
 - (a) diagnose capacity and training length
 - (b) record 0.99 as out of scope
 - Recommend (a); a sweep cannot separate the causes.
- **D3 Read held-out data yet? Recommend not yet; high confidence.**

Proposed next step

- **On D1(a): repeat the Ising budget-6 cell with fresh generation seeds.**
- **On D2(a): write the capacity-versus-search protocol and its cap, then stop.**
- **Either yes commits to a new protocol and cap before anything runs.**
- **Already committed: report, hashed logs, reproduction commands.**

Recommendation and decision are recorded separately in the research log; nothing above is approved.